H2 Math Summary

Notes

Maximus Pung

Contents

1 Graphing Techniques **2**

1.1 Curve Sketching . . . 2
1.1.1 Vertical Asymptotes . . . 2
1.1.2 Horizontal Asymptotes . . . 3
1.1.3 Oblique Asymptotes . . . 3
1.2 Graphs of Rational Functions . . . 5
1.3 Symmetry . . . 6
1.4 Basic Graphs . . . 7
1.5 Parametric Equations . . . 8
1.5.1 Finding the Cartesian Equation . . . 8
1.5.2 Asymptotes . . . 9
1.5.3 Intersections . . . 9
1.6 Conics . . . 10

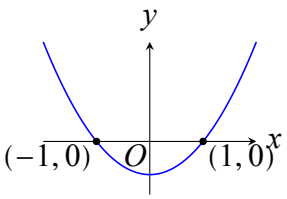
1 Graphing Techniques

1.1 Curve Sketching

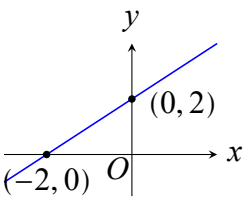
Key features of a curve:

(a) **Axial intercepts**, when $x=0$ or $y=0$.

x-intercept: set $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.

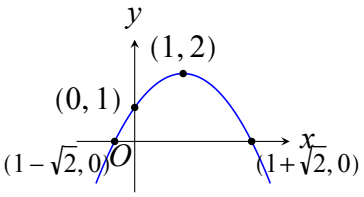


y-intercept: set $x=0$.
e.g. $y=x+2$: $y=2$.

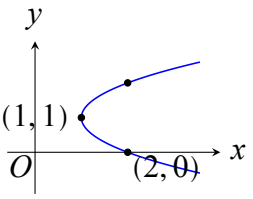


(b) **Stationary points**, when $\frac{dy}{dx}=0$ or $\frac{dy}{dx}$ is undefined. Label upon request.

Horizontal turning point: $\frac{dy}{dx}=0$ (tangent horizontal).
e.g. $y=-(x-1)^2+2$, maximum point $(1, 2)$.



Vertical turning point: $\frac{dy}{dx}$ undefined (tangent vertical).
e.g. $x=(y-1)^2+1$, turning point $(1, 1)$.



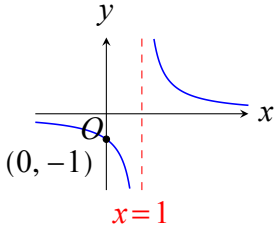
(c) **Asymptotes**

1.1.1 Vertical Asymptotes

If as $x \to k$, $y \to \pm\infty$, then $x=k$ is a vertical asymptote. Common occurrences:

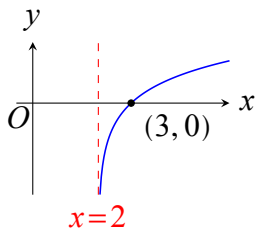
1. Fractions. Denominator $=0$.

e.g. $y=\frac{1}{x-1}$: vertical asymptote $x=1$.



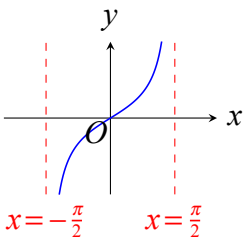
2. Logarithmic graphs. Inside $\ln = 0$.

e.g. $y = \ln(x-2)$: vertical asymptote $x = 2$.



3. Tan graphs. Inside $\tan = \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$.

e.g. $y = \tan x$: vertical asymptotes $x = \pm\frac{\pi}{2},\ \pm\frac{3\pi}{2}, \ldots$



1.1.2 Horizontal Asymptotes

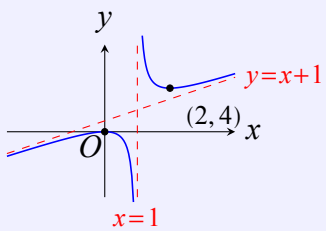
As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

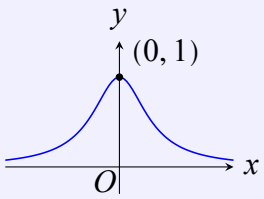
Useful trick to find horizontal asymptotes

For rational fns $\frac{P(x)}{Q(x)}$.

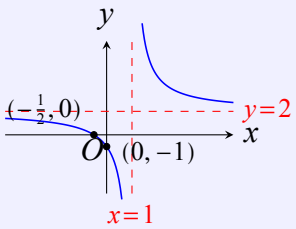
numerator power > denom power
⇒ no horizontal asymptote
e.g. $y = \frac{x^2}{x-1}$



numerator power < denom power
⇒ horiz asymptote: $y = 0$
e.g. $y = \frac{1}{x^2+1}$



numerator power = denom power
⇒ horiz asymptote: ratio of coeff of highest power.
e.g. $y = \frac{2x+1}{x-1},\ y = 2$



1.1.3 Oblique Asymptotes

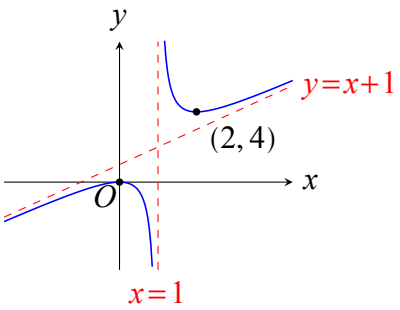
An asymptote of the form $y = mx + c,\ m \neq 0$.

e.g. $y=2x+1+\dfrac{3}{x-2}$, $y=3x-5+e^{-x}$

$$y=\frac{ax^2+bx+c}{dx+e}=Ax+B+\frac{C}{dx+e}$$

Oblique asymptote: $y=Ax+B$. Vertical asymptote: $x=-\dfrac{e}{d}$.

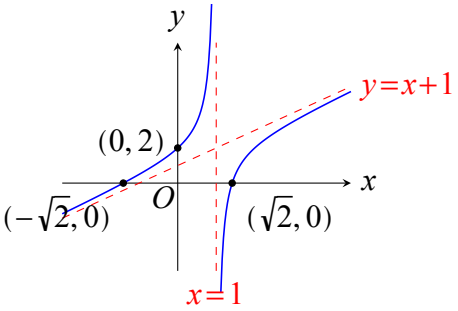
2 turning points



e.g. $y=x+1+\dfrac{1}{x-1}$

$\left(\dfrac{Cd}{A}>0\right)$

No turning points.



e.g. $y=x+1-\dfrac{1}{x-1}$

$\left(\dfrac{Cd}{A}<0\right)$

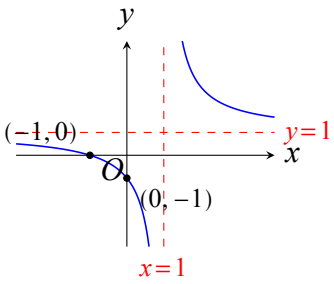
Note

For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian equation.

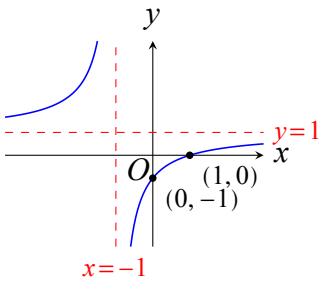
1.2 Graphs of Rational Functions

Two common shapes for each type.

Linear / linear: $y=\dfrac{Ax+B}{Cx+D}$ Horizontal asymptote: $y=\dfrac{A}{C}$. Vertical asymptote: $x=-\dfrac{D}{C}$.

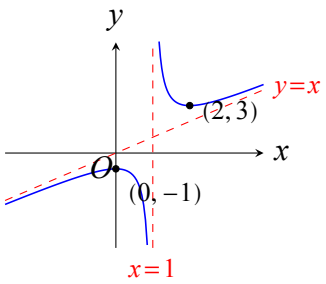


e.g. $y=\dfrac{x+1}{x-1}$

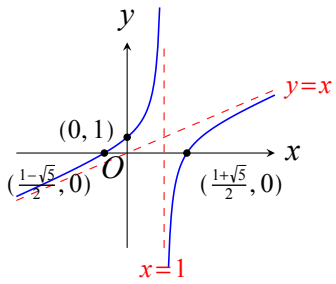


e.g. $y=\dfrac{x-1}{x+1}$

Quadratic / linear: $y=\dfrac{Ax^2+Bx+C}{Dx+E}$ Oblique asymptote; vertical asymptote $x=-\dfrac{E}{D}$.

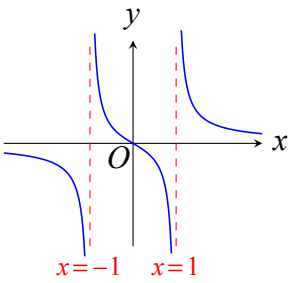


e.g. $y=x+\dfrac{1}{x-1}$

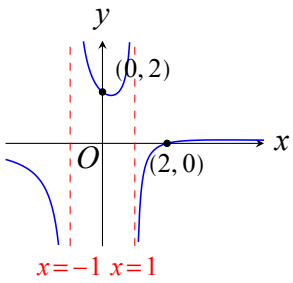


e.g. $y=x-\dfrac{1}{x-1}$

Linear / quadratic: $y=\dfrac{ax+b}{cx^2+dx+e}$ Horizontal asymptote: $y=0$. Vertical asymptotes: Solve $cx^2+dx+e=0$ (possibly no roots).



e.g. $y=\dfrac{x}{(x-1)(x+1)}$



e.g. $y=\dfrac{x-2}{(x-1)(x+1)}$

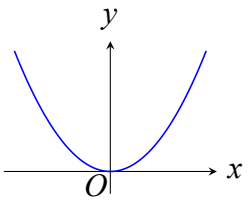
Quadratic curves: $y=a(x-h)^2+k$, $y=\alpha(x-m)(x-n)$.

1.3 Symmetry

Even functions: $\mathrm{f}(-x)=\mathrm{f}(x)$.

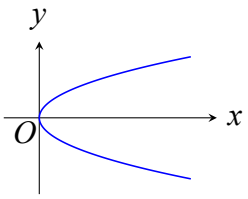
The graph is symmetric about the y-axis.

e.g. $y=x^2$



Symmetry about the x-axis: replacing y by $-y$ gives the same equation.

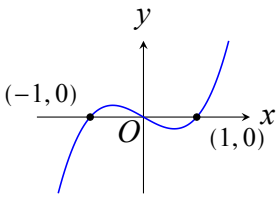
e.g. $y^2=x$



Odd functions: $\mathrm{f}(-x)=-\mathrm{f}(x)$.

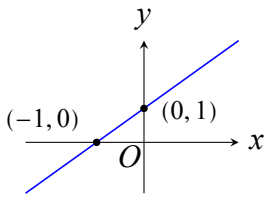
The graph is symmetric about the origin.

e.g. $y=x^3-x$

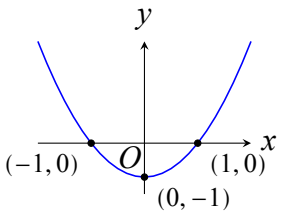


1.4 Basic Graphs

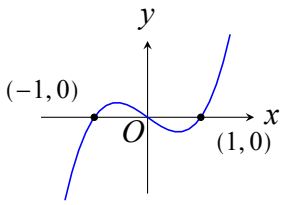
- Polynomial (linear, quadratic, cubic).
- Exponential. $y=ae^x+b$.
- Logarithm. $y=\ln(ax+b)$.
- Trigo. $y=\sin x$, $\cos x$, $\tan x$.
- Inverse Trigo. $y=\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$.



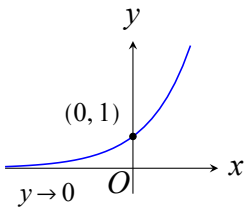
Linear $y=x+1$



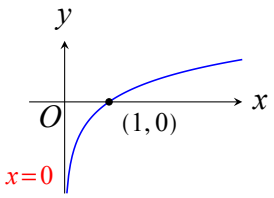
Quadratic $y=x^2-1$



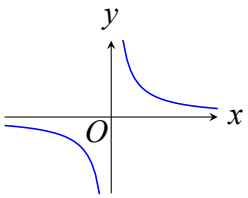
Cubic $y=x^3-x$



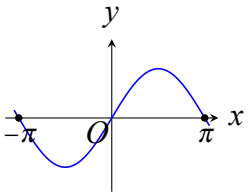
Exponential $y=e^x$



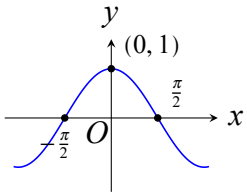
Logarithm $y=\ln x$



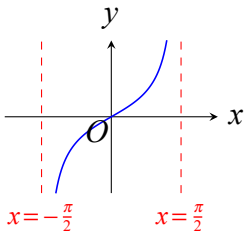
Reciprocal $y=\frac{1}{x}$



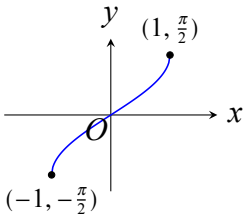
$y=\sin x$



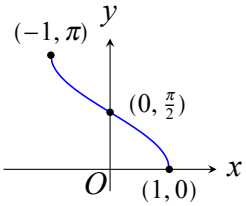
$y=\cos x$



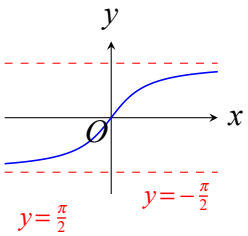
$y=\tan x$



$y=\sin^{-1}x$



$y=\cos^{-1}x$



$y=\tan^{-1}x$

1.5 Parametric Equations

A curve C can be described by giving both coordinates in terms of a parameter t:

$$x=x(t), \quad y=y(t), \quad t\in T$$

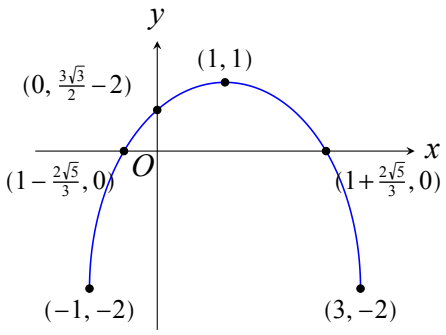
Example.

$x=2\cos t+1, \quad y=3\sin t-2, \quad 0\leqslant t\leqslant \pi$

$t=0$: $(3,-2)$ $\quad t=\frac{\pi}{2}$: $(1,1)$ $\quad t=\pi$: $(-1,-2)$

$x=0$: $\left(0, \frac{3\sqrt{3}}{2}-2\right)$

$y=0$: $\left(1\pm\frac{2\sqrt{5}}{3}, 0\right)$



CAUTION: sketching from parametric equations

ALWAYS take note of the range of t when sketching graphs. The restriction on t decides which part of the curve exists: above, $0\leqslant t\leqslant\pi$ gives only the upper half of the ellipse, with endpoints $(3,-2)$ and $(-1,-2)$, not the whole ellipse.

1.5.1 Finding the Cartesian Equation

The cartesian equation of C is an equation without the parameter t. Common techniques:

① **Make t the subject.** e.g. $x=2t-1, y=3t^2+\mathrm{e}^t, t\in\mathbb{R}$

$$t=\frac{x+1}{2}$$

$$y=3\left(\frac{x+1}{2}\right)^2+\mathrm{e}^{\frac{x+1}{2}}$$

② **Trigo identity formulas.** e.g. $x=2\cos t+1, y=3\sin t-2, 0\leqslant t\leqslant\pi$

$$\cos t=\frac{x-1}{2}, \quad \sin t=\frac{y+2}{3}$$

$$\cos^2 t+\sin^2 t=1$$

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{9}=1, \quad y\geqslant -2$$

③ **Others.** e.g. $x=t-\frac{1}{t}, y=t+\frac{1}{t}, t\neq 0$

$$y+x=2t, \quad y-x=\frac{2}{t}$$

$$(y+x)(y-x)=4$$

$$y^2-x^2=4$$

Note: Avoid finding the cartesian eqn if the question did not ask you to.

1.5.2 Asymptotes

- Find (if possible) $t \to t_0$ such that $x \to x_0, y \to \pm\infty$ (vertical asymptote $x = x_0$), or $y \to y_0, x \to \pm\infty$ (horizontal asymptote $y = y_0$).
- Oblique asymptotes (if they exist) can only be found via the cartesian equation.

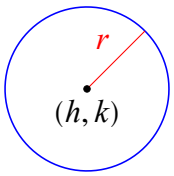
1.5.3 Intersections

Intersection between a parametric curve and another curve: sometimes both equations are needed.

1.6 Conics

Circle: $(x-h)^2+(y-k)^2=r^2$

Centre: (h,k) radius: r

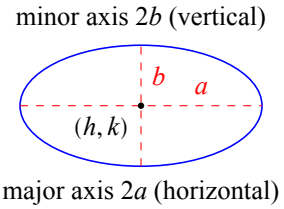


Ellipse: $\dfrac{(x-h)^2}{a^2}+\dfrac{(y-k)^2}{b^2}=1$

Centre: (h,k), "horizontal radius" $=a$, "vertical radius" $=b$

(for $a>b$: major axis horizontal, length $2a$, semi-major a; minor axis vertical, length $2b$, semi-minor b)

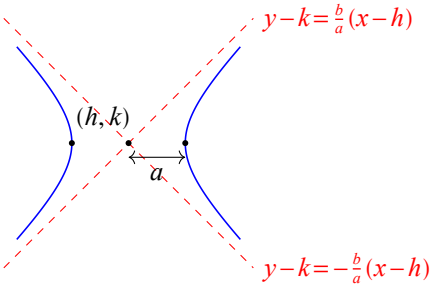
⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.



Hyperbola: Type I

$\dfrac{(x-h)^2}{a^2}-\dfrac{(y-k)^2}{b^2}=1$

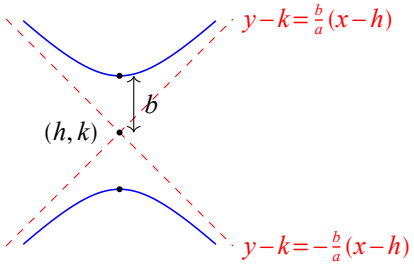
Centre: (h,k), Distance from centre to vertices $=a$



Type II

$\dfrac{(y-k)^2}{b^2}-\dfrac{(x-h)^2}{a^2}=1$

Centre: (h,k), Distance from centre to vertices $=b$



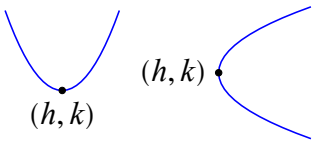
⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.

⋆ Equation of asymptotes: $y-k=\pm\dfrac{b}{a}(x-h)$ (dashed in the sketches).

Parabola:

· $y-k=a(x-h)^2$ Turning point $\left(\dfrac{dy}{dx}=0\right)$: (h,k)

· $x-h=b(y-k)^2$ Turning point $\left(\dfrac{dy}{dx}\text{ undefined}\right)$: (h,k)



Common question types for curve sketching

→ Find the range/set of values that y can/cannot take. [VJC 2025]

→ Find the range of values of a such that C has no turning points [SAJC 2025]

→ Line rotation about pivot question. [SAJC 2025]

→ Conic superimposed onto graph. [DHS 2025]